Divya Appapogu

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Research Interests: I am broadly interested in interpretability and understanding language models at a fundamental level. My current work focuses on the studying geometry of internal representations of language models, with the goal of understanding how they represent concepts.

Education

Ph.D. in Computer Science, Boston University — Sept 2024 – Present
Advised by Dr. Aaron Mueller

M.S. in Artificial Intelligence, Boston University — Aug 2021 – Jan 2023
Advised by Dr. Bryan Plummer

B.Tech in Engineering Science, IIT Hyderabad — Aug 2015 – May 2019
Advised by Dr. Soumya Jana

Publications

Preprints

MAxBench: A Multinomial Concept Recovery Benchmark [PDF] [Code]
D. Appapogu, F. Behrens, Y. Belinkov, A. Mueller. **arXiv preprint 2026**.

Multi-axis Analysis of Image Manipulation Localization [PDF]
K. Nichols, D. Appapogu, G. Biamby, D. Bashkirova, A. Rohrbach, B. A. Plummer. **arXiv preprint 2026**.

Workshop Papers

Localizing and Evaluating Multinomial Concepts with Affine Subspaces [PDF]
D. Appapogu, F. Behrens, Y. Belinkov, A. Mueller. **ICML 2026 Mechanistic Interpretability Workshop**.

Journal Articles

Anatomy-Guided, Modality-Agnostic Segmentation of Neuroimaging Abnormalities [PDF] [Code]
D. Lteif, D. Appapogu, S. A. Bargal, B. A. Plummer, V. B. Kolachalama. **Human Brain Mapping 2025**.

Industry Experience

Senior Software Engineer, **Barclays**, New Jersey — Feb 2023 – Aug 2024
Built scalable backend services (Java, Spring Boot, Kafka) deployed via Docker/OpenShift CI/CD.

Senior Software Engineer, **OYO**, Bangalore, India — June 2019 – Aug 2021
Led monolith-to-microservices migration; built Kafka data pipelines and optimized search and caching at scale.

Invited Talks

Localizing and Evaluating Multinomial Concepts with Affine Subspaces, AIR Seminar, Boston University — 2025

Teaching

Teaching Assistant, CS506 — Tools for Data Science
Boston University · Fall 2025

Teaching Assistant, CS101 — Introduction to Computing
Boston University · Fall 2021, Spring 2022, Fall 2024

Service

Student Organizer — New England Mechanistic Interpretability (NEMI) Workshop hosted at Boston University, 2026 — 2026

Skills

Research: Mechanistic Interpretability, Geometry of Language Models, Medical Imaging, Linear Algebra, Probability

Technical: Python, PyTorch, Hugging Face Transformers, nnsight, TransformerLens, NumPy, scikit-learn, C++, Java, SQL